

Does science need rescuing?



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"Whatever the scientific community decides, we shouldn't forget that science is unique amongst all pursuits of the human race"

In a time witnessing rapid technological advancements, we reap the rewards of yesteryear's scientific breakthroughs. There's a growing interest in science now than ever before, both in advanced as well as developing nations of the world. Merriam-Webster's online dictionary saw a 176% spike in search terms for the word "science" three years ago. According to Google, more people from countries such as India, Ghana, Nigeria, Philippines and other developing economies seem to be interested in the definition of science online. There was never a better time in history to be a science nerd than now, especially in the West (or developed economies) where scientific and technological breakthroughs are seeping into the larger cultural consciousness.

Just look at 2015 – events such as the Pluto fly-by, irrefutable evidence of water on Mars, and SpaceX's historic controlled landing of a rocket returning from space not only grabbed international headlines but led to a larger global dialogue thanks to memes, jokes and a broader social commentary. All these things wouldn't have been possible without years of painstaking scientific enquiry.

But is there a point beyond which scientific enquiry itself begins to lose its meaning? Some new and popular scientific theories (yes String Theory, I mean you), just aren't testable, don't make real predictions, and aren't even falsifiable. Don't get me wrong, there's nothing wrong with taking leaps of imagination, even in science. Einstein did thought experiments, and ended up giving us the most beautiful scientific advances since Newton! However, when theories have been around for 40 years and still can't make predictions, still are purely mathematically based with no grounding in testable reality... well... people are going to start to question it then.

This is exactly what happened in December 2015. I read an interesting article online (<http://digit.in/ScienceDebate>). Some of the world's foremost scientific thinkers and particle physicists assembled in a Munich university hall for a three-day conference debating the need to tackle a growing divide in the scientific community at a higher philosophical level. The questions they tried to answer were quite fundamental: What is science and what isn't? And should progress cease simply because scientific theories haven't been proven yet?

The answer to both these questions, as you would've guessed, is complicated.

At the Munich debate, the merits and demerits of overindulging in the current grand unification

theory of all particulate matter was questioned and weighed. As always, a recurring bone of contention within the scientific community since its proposition, string theory continued to divide camps. According to its detractors not a single piece of verifiable evidence has been gathered to support the scientific validity of string theory, as almost all of it is just mathematical elegance with virtually no physically justifiable component – yet. What's more, physicists eager to leave their mark are trying to come up with new theories built on top of string theory, with the same fundamental problem. Castles in the air built on top of other castles in the air?

Perhaps it's time to start finding better labels for theories that can never be proven or disproven in the foreseeable future. Maybe they should be meta-sciences, or non-science until they find a way of predicting something measurable (which would also give us the ability to prove or disprove them).

We're also starting to hit natural boundaries. The LHC and other particle accelerators are hitting the detection boundaries of the very small. Our most advanced telescopes have hit a physical barrier of how far into space (and the past) we can look. We're literally held captive by ignorance, because we can't seem match the universe's highest and lowest magnification levels! Perhaps then, this needs us to take a few flights of fancy in order to stay at the cutting edge of understanding, but then it also opens up many chinks in the armour of science, which many others are quick to exploit, and then use that to disregard science altogether.

This is exactly why the discussions in Munich were of utmost importance. Whether theoretical physicists and experimental physicists can find an uncomfortable (but necessary) middle ground to co-exist, we shall see. This would at least give us a basic definition of what "science" really is.

Whatever the scientific community decides, we shouldn't forget that science is unique amongst all pursuits of the human race. The sanctity of science needs to be preserved beyond everything else, we cannot let it get maligned at any cost. It's up to modern and futuristic technology to step up its game and rescue science. Who will rise to the occasion? 

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